Containerization-and-DevOps

🐳 Class Task

🛠️ Tools Used

- Docker
- Nginx
- WSL

⚙️ Steps

Run Nginx Container with Bind Mount

```
docker run -d \
--name nginx-container \
--mount type=bind,source=/home/sristee/html,target=/usr/share/nginx/html \
-p 8080:80 \
nginx
```

Remove Container (Incorrect Command Attempt)

```
docker rm -f nginx container
```

Error:

- No such container: nginx
- No such container: container

Remove Container (Correct Command)

```
docker rm -f nginx-container
```

Run Container with tmpfs Mount

```
docker run -d \
--name temp-container \
--mount type=tmpfs,target=/app/cache \
nginx
```

⚠️ Notes

- Bind mount links host directory to container (`/home/sristee/html`)
- Incorrect container name caused removal error
- `tmpfs` mount stores data in memory (non-persistent)
- Data in `tmpfs` is lost when container stops

✅ Outcome

- Successfully used bind mount with Nginx
- Understood container naming importance
- Implemented `tmpfs` mount
- Learned difference between persistent and non-persistent storage

📸 Screenshots

```
sristee@Sristee:/mnt/c/Users/srist$ docker run -d \
  --name nginx-container \
  --mount type=bind,source=/home/sristee/html,target=/usr/share/nginx/html \
  -p 8080:80 \
  nginx
6f6e4d1bb5cc31dd6f02ca9894da37fc53671f1f3390cf2a34a35f6f541866ed
sristee@Sristee:/mnt/c/Users/srist$ docker rm -f nginx container
Error response from daemon: No such container: nginx
Error response from daemon: No such container: container
sristee@Sristee:/mnt/c/Users/srist$ docker rm -f nginx-container
nginx-container
sristee@Sristee:/mnt/c/Users/srist$ docker run -d \
  --name temp-container \
  --mount type=tmpfs,target=/app/cache \
  nginx
2f029def77725e025d4cc6d8759443741c640db5b9197b4c8ed29f4bc4e15f35
sristee@Sristee:/mnt/c/Users/srist$
```